Research related to Artificial Intelligence

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May 13th Hochschule Darmstadt

Today’s menu

Data Science /
engineering

- large project software engineering (4 years)
- full stack, distributed web application
- international (+10 members, English, Spanish, German)
- agile, test driven development
- data curation for Artificial Intelligence

Protein function prediction /
applied bioinformatics using lexical analysis

- efficient, high throughput tool
- annotation based on lexical analysis

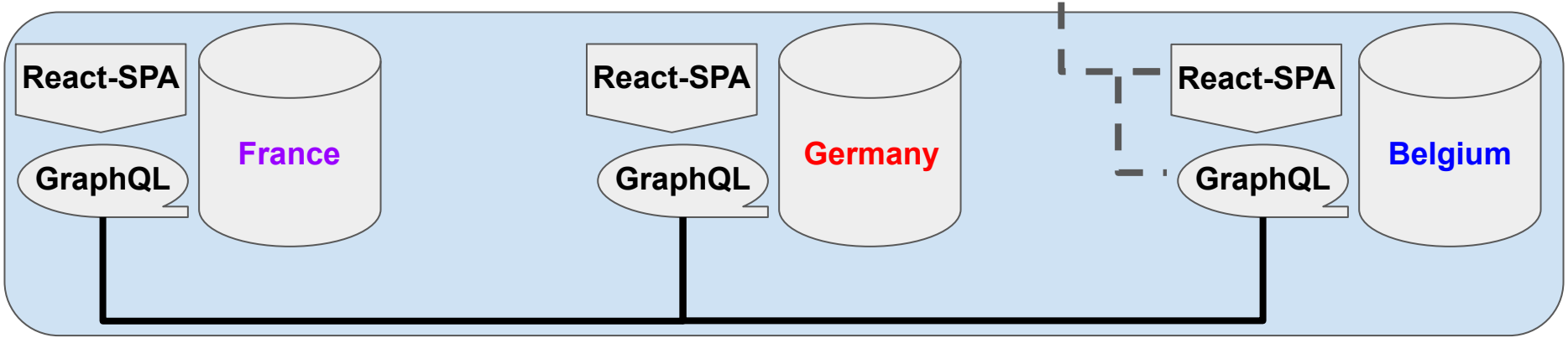
facebook.com

1. Automated data-warehouse generation

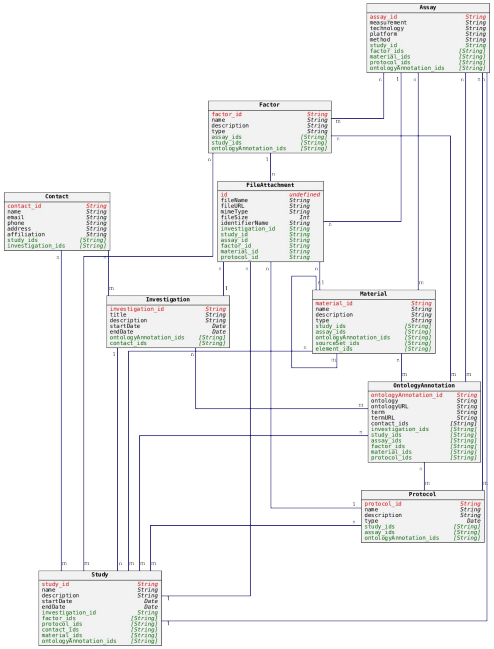
- Data standardization
- Data integration
- Data FAIR-ification

endr.eu; wikimedia.org

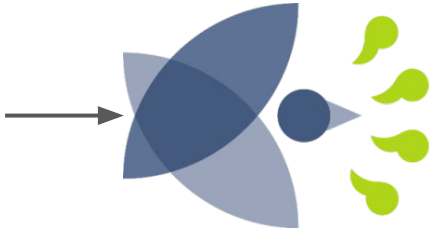
1. Data-Cloud of independent servers



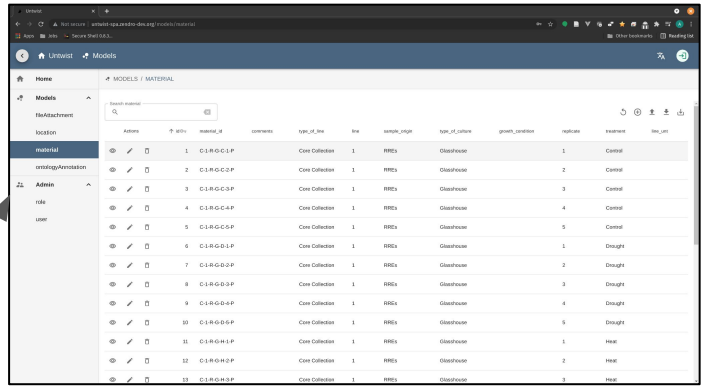
1. Zendro: Code generators



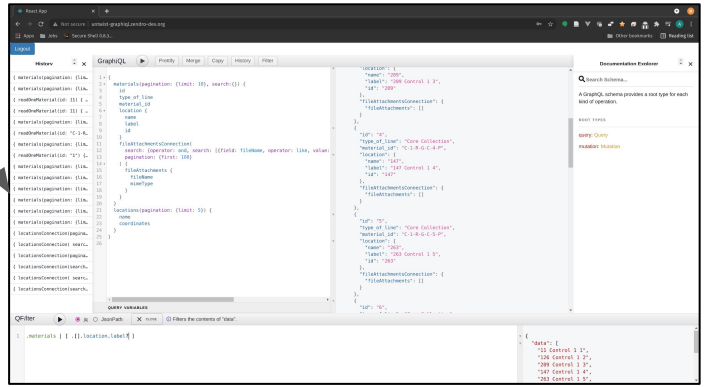
Data-Model Definitions



Zendro



Browser based interface - React.js



Programmatic API - GraphQL

1. Zendro Data Models

```
{
  "model": "location",
  "storageType": "sql",
  "attributes": {
    "name": "String",
    "label": "String",
    "type": "String",
    "coordinates": "[Int]",
    "bounding_box": "[Int]",
    "abidance_start_date": "DateTime",
    "abidance_end_date": "DateTime",
    "parent_id": "Int",
    "root_id": "Int",
    "material_id": "Int"
  },
  "associations": {
    "material": {
      "type": "one_to_one",
      "implementation": "foreignkeys",
      "reverseAssociation": "location",
      "targetStorageType": "sql",
      "target": "material",
      "targetKey": "material_id",
      "keysIn": "location",
      "label": "material_id"
    },
    "parent": {
      "type": "many_to_one",
      "implementation": "foreignkeys",
      "reverseAssociation": "descendants",
      "target": "location",
      "targetKey": "parent_id",
      "keysIn": "location",
      "targetStorageType": "sql",
      "label": "name"
    },
    "descendants" : {
      "type": "one_to_many",
      "implementation": "foreignkeys",
```

Define each data model

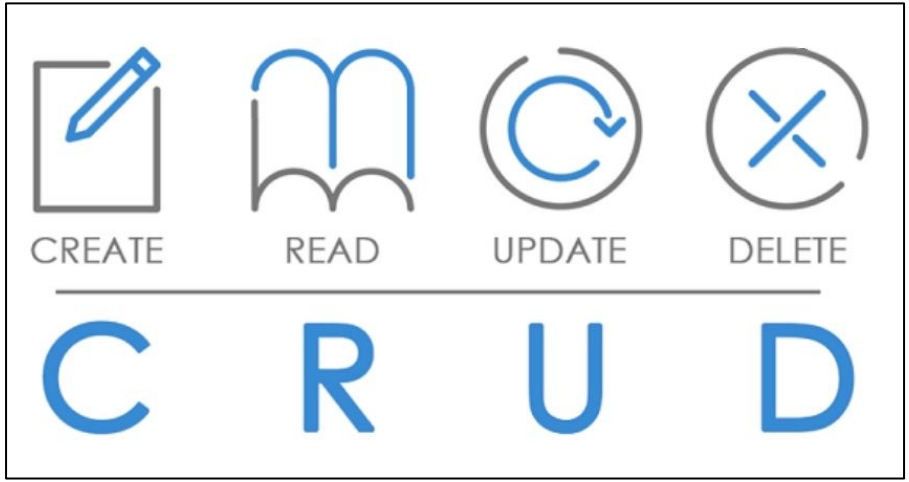
- name
- attributes
- associations
- storage-technology:

1. Zendro: Automatically generated CRUD-API

- ***C***reate
- ***R***ead including search
- ***U***pdate
- ***D***elete

and

- exhaustive searches across
  - relations, spread over
  - databases and / or
  - servers



betalabs.in

Operators:

1. Zendro: Full predicate logic search expressions

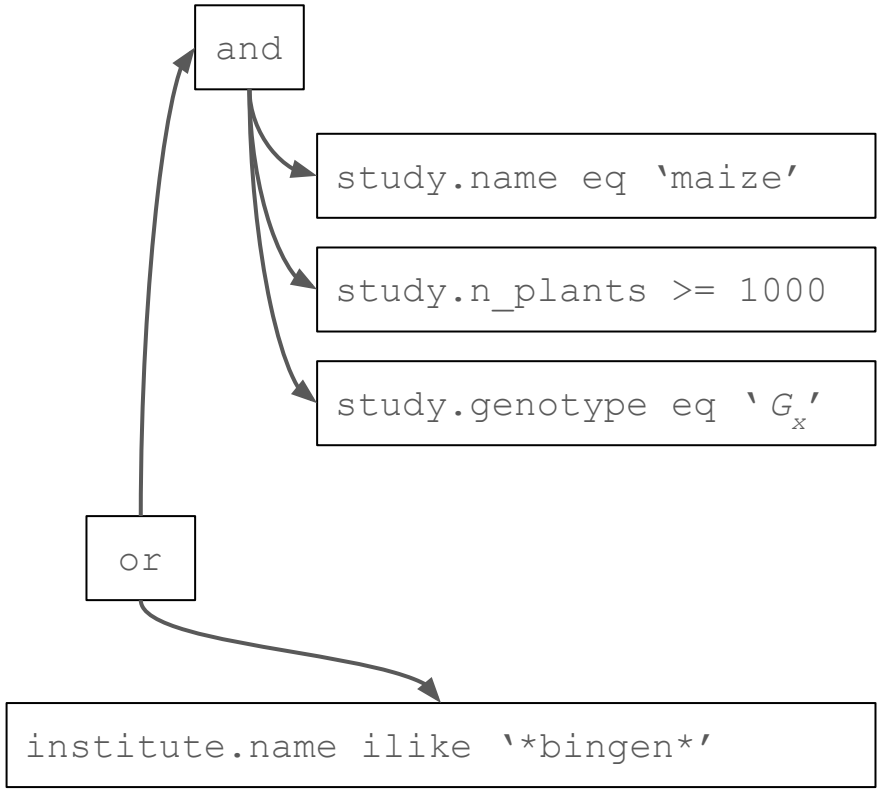
Exhaustive searches:

- any complex selection criteria, e.g.

  "Drought response on maize using genotype G_x with protocol P_i and at least 1000 studied plants."

  extend:

  "*... or institute TH Bingen.*"



like
notLike
iLike
notILike
regexp
notRegexp
iRegexp
notIRegexp
eq
gt
gte
lt
lte
ne
between
notBetween
in
notIn
contains
notContains
or
and
not

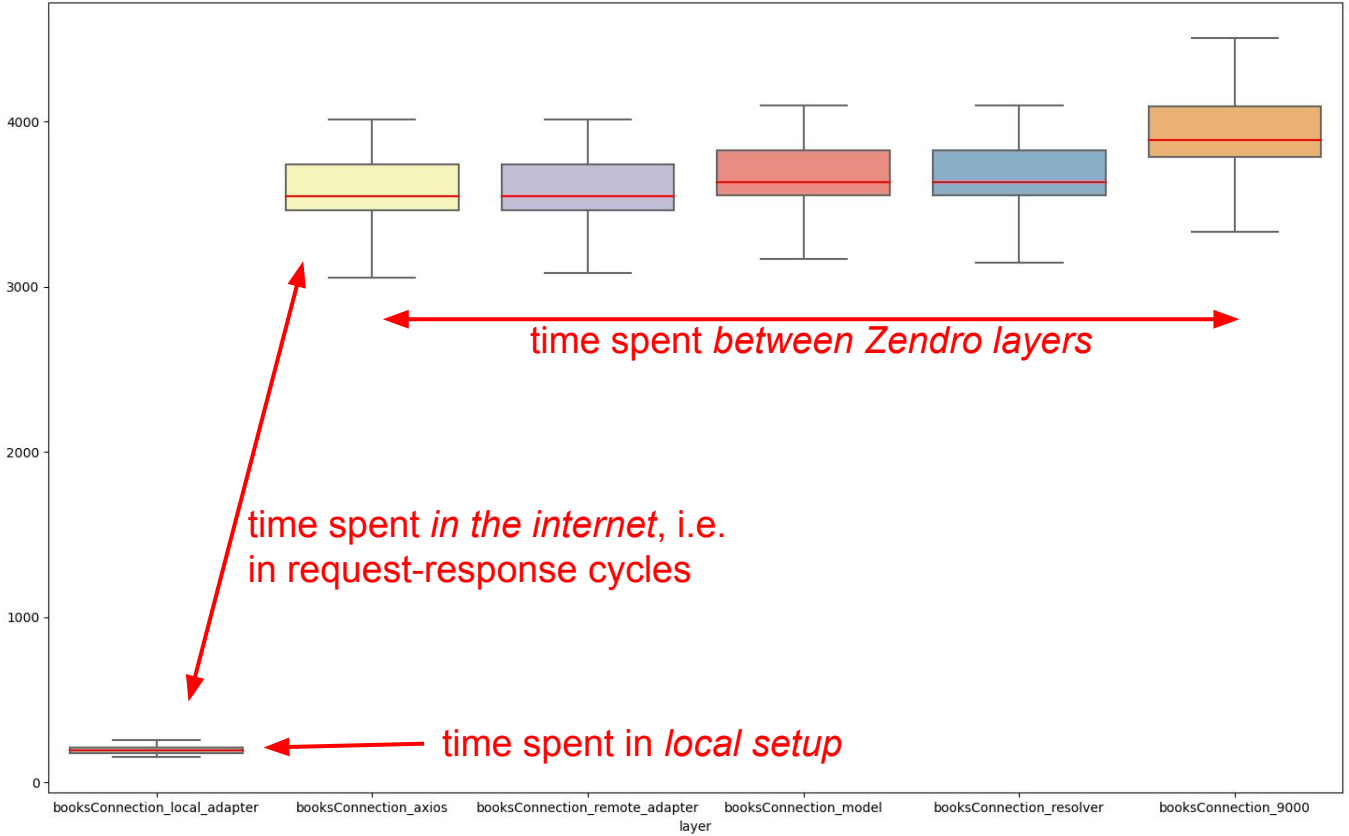
1. Zendro: Distributed efficiency, time in ms

Distribute data models over

- storages
- servers

Limiting factor:

Connection speed of slowest node!



1. Repository

zendro-dev.github.io

1. Research projects using Zendro

- EMPHASIS
- EPPN2020
- EOSC-Life
- Reconstruct
- BreedPatH
- DataPlant
- Teosinte

Bundesministerium
für Bildung
und Forschung

2. Previous results

Build on results from previous work (PhD thesis, Tomato genome project [1]).

<u>Classification Problem:</u>
Provide short descriptions of the molecular functions to chemical formulae of proteins.

Prototype successfully used for Tomato proteins.

nature.com

[1] The tomato genome sequence provides insights into fleshy fruit evolution. Nature 485, 635–641 (2012)

2. Protein function prediction: “Prot.Scriber”

Assign short and concise human readable descriptions to query proteins.

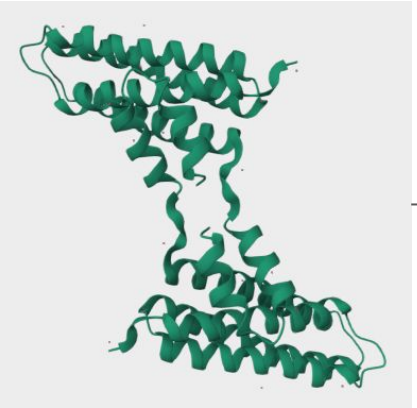
```
>AT1G01040.1 | dicer-like 1 | Chr1:23519-31079 FORWARD LENGTH=1909 | 201606
>AT1G01040.2 | dicer-like 1 | Chr1:23519-31079 FORWARD LENGTH=1910 | 201606
>AT1G01050.1 | pyrophosphorylase 1 | Chr1:31382-32670 REVERSE LENGTH=212 | 201606
>AT1G01050.2 | pyrophosphorylase 1 | Chr1:31382-32670 REVERSE LENGTH=212 | 201606
>AT1G01060.1 | Homeodomain-like superfamily protein | Chr1:33992-37061 REVERSE LENGTH=645 | 201606
>AT1G01060.2 | Homeodomain-like superfamily protein | Chr1:33992-37061 REVERSE LENGTH=645 | 201606
>AT1G01060.3 | Homeodomain-like superfamily protein | Chr1:33992-37061 REVERSE LENGTH=645 | 201606
>AT1G01060.4 | Homeodomain-like superfamily protein | Chr1:33992-37061 REVERSE LENGTH=644 | 201606
>AT1G01060.5 | Homeodomain-like superfamily protein | Chr1:33992-36836 REVERSE LENGTH=643 | 201606
>AT1G01060.6 | Homeodomain-like superfamily protein | Chr1:33992-36836 REVERSE LENGTH=643 | 201606
>AT1G01060.7 | Homeodomain-like superfamily protein | Chr1:33992-37061 REVERSE LENGTH=645 | 201606
>AT1G01060.8 | Homeodomain-like superfamily protein | Chr1:33992-37061 REVERSE LENGTH=645 | 201606
>AT1G01070.1 | nodulin MtN21 /EamA-like transporter family protein | Chr1:38898-40877 REVERSE LENGTH=365 | 201606
>AT1G01070.2 | nodulin MtN21 /EamA-like transporter family protein | Chr1:38898-40597 REVERSE LENGTH=318 | 201606
>AT1G01080.1 | RNA-binding (RRM/RBD/RNP motifs) family protein | Chr1:45503-46789 REVERSE LENGTH=293 | 201606
>AT1G01080.2 | RNA-binding (RRM/RBD/RNP motifs) family protein | Chr1:45503-46789 REVERSE LENGTH=294 | 201606
>AT1G01080.3 | RNA-binding (RRM/RBD/RNP motifs) family protein | Chr1:45610-46789 REVERSE LENGTH=286 | 201606
>AT1G01090.1 | pyruvate dehydrogenase E1 alpha | Chr1:47705-49166 REVERSE LENGTH=428 | 201606
>AT1G01100.1 | 60S acidic ribosomal protein family | Chr1:50284-50954 REVERSE LENGTH=112 | 201606
>AT1G01100.2 | 60S acidic ribosomal protein family | Chr1:50284-50954 REVERSE LENGTH=112 | 201606
>AT1G01100.3 | 60S acidic ribosomal protein family | Chr1:50284-50954 REVERSE LENGTH=96 | 201606
>AT1G01100.4 | 60S acidic ribosomal protein family | Chr1:50284-50954 REVERSE LENGTH=112 | 201606
>AT1G01110.1 | IQ-domain 18 | Chr1:53022-54494 FORWARD LENGTH=364 | 201606
>AT1G01110.2 | IQ-domain 18 | Chr1:52239-54494 FORWARD LENGTH=527 | 201606
>AT1G01120.1 | 3-ketoacyl-CoA synthase 1 | Chr1:57392-58978 REVERSE LENGTH=528 | 201606
>AT1G01130.1 | CBL-interacting Serine/Threonine-kinase | Chr1:61905-63811 REVERSE LENGTH=124 | 201606
>AT1G01140.1 | CBL-interacting protein kinase 9 | Chr1:64398-67512 REVERSE LENGTH=447 | 201606
>AT1G01140.2 | CBL-interacting protein kinase 9 | Chr1:64398-67512 REVERSE LENGTH=449 | 201606
>AT1G01140.3 | CBL-interacting protein kinase 9 | Chr1:64398-67512 REVERSE LENGTH=451 | 201606
>AT1G01150.1 | Homeodomain-like protein with RING/FYVE/PHD-type zinc finger domain-containing protein | Chr1:70115-7
>AT1G01150.2 | Homeodomain-like protein with RING/FYVE/PHD-type zinc finger domain-containing protein | Chr1:70115-7
>AT1G01150.3 | Homeodomain-like protein with RING/FYVE/PHD-type zinc finger domain-containing protein | Chr1:70828-7
>AT1G01160.1 | GRF1-interacting factor 2 | Chr1:72583-73883 FORWARD LENGTH=195 | 201606
>AT1G01160.2 | GRF1-interacting factor 2 | Chr1:72583-73883 FORWARD LENGTH=229 | 201606
>AT1G01170.1 | ozone-responsive stress-like protein (DUF1138) | Chr1:74105-74443 REVERSE LENGTH=83 | 201606
>AT1G01170.2 | ozone-responsive stress-like protein (DUF1138) | Chr1:74105-74443 REVERSE LENGTH=83 | 201606
>AT1G01180.1 | S-adenosyl-L-methionine-dependent methyltransferases superfamily protein | Chr1:75633-76556 FORWARD L
>AT1G01190.1 | cytochrome P450%2C family 78%2C subfamily A%2C polypeptide 8 | Chr1:83045-84864 REVERSE LENGTH=535 |
>AT1G01190.2 | cytochrome P450%2C family 78%2C subfamily A%2C polypeptide 8 | Chr1:83045-84879 REVERSE LENGTH=540 |
>AT1G01200.1 | RAB GTPase homolog A3 | Chr1:86715-88145 REVERSE LENGTH=237 | 201606
>AT1G01210.1 | DNA-directed RNA polymerase%2C subunit M%2C archaeal | Chr1:88977-89529 FORWARD LENGTH=106 | 201606
```

Araport11 proteins

2. Generate input

Sequence Similarity Searches:

Compare chemical structures of query proteins with references



BLAST

Basic Local Alignment Search Tool

```
>SP:Q9LNF2 PMEI1_ARATH Pectinesterase inhibitor 1 OS=Arabidopsis thaliana
OX=3702 GN=PMEI1 PE=1 SV=1
Length=176

 Score = 342 bits (878),  Expect = 2e-121
 Identities = 176/176 (100%), Positives = 176/176 (100%), Gaps = 0/176 (0%)

Query  1    MAANLRNNAFLSSLMFLLLIGSSYAITSSEMSTICDKTLNPSFCLKFLNTKFASPNLQAL  60
            MAANLRNNAFLSSLMFLLLIGSSYAITSSEMSTICDKTLNPSFCLKFLNTKFASPNLQAL
Sbjct  1    MAANLRNNAFLSSLMFLLLIGSSYAITSSEMSTICDKTLNPSFCLKFLNTKFASPNLQAL  60

Query  61   AKTTLDSTQARATQTLKKLQSIIDGGVDPRSKLAYRSCVDEYESAIGNLEEAFEHLASGD  120
            AKTTLDSTQARATQTLKKLQSIIDGGVDPRSKLAYRSCVDEYESAIGNLEEAFEHLASGD
Sbjct  61   AKTTLDSTQARATQTLKKLQSIIDGGVDPRSKLAYRSCVDEYESAIGNLEEAFEHLASGD  120

Query  121  GMGMNMKVSAALDGADTCLDDVKRLRSVDSSVVNNSKTIKNLCGIALVISNMLPRN  176
            GMGMNMKVSAALDGADTCLDDVKRLRSVDSSVVNNSKTIKNLCGIALVISNMLPRN
Sbjct  121  GMGMNMKVSAALDGADTCLDDVKRLRSVDSSVVNNSKTIKNLCGIALVISNMLPRN  176


>SP:Q9LUV1 PMEI2_ARATH Pectinesterase inhibitor 2 OS=Arabidopsis thaliana
OX=3702 GN=PMEI2 PE=1 SV=1
Length=173

 Score = 144 bits (363),  Expect = 4e-43
 Identities = 79/175 (45%), Positives = 109/175 (62%), Gaps = 4/175 (2%)

Query  1    MAANLRNNAFLSSLMFLLLIGSSYAITSSEMSTICDKTLNPSFCLKFL--NTKFASPNLQ  58
            MAA L N   +SSLMF ++ GS  A  + ++  IC K  N SFC  ++  N K +  +LQ
Sbjct  1    MAAYLTNRVLMSSLMFFVMTGSLNAQVA-DIKAICGKAKNQSFCTSYMKSNPKTSGADLQ  59

Query  59   ALAKTTLDSTQARATQTLKKLQSIIDGGVDPRSKLAYRSCVDEYESAIGNLEEAFEHLAS  118
             LA  T  S Q  A++  +K+QS++    +P  K AY SCV  Y+SAI +L +A + LAS
Sbjct  60   TLANITFGSAQTSASEGFRKIQSLVKTATNPTMKKAYTSCVQHYKSAISSLNDAKQSLAS  119

Query  119  GDGMGMNMMKVSAALDGADTCLDDVKRLRSVDSSVVNNSKTIKNLCGIALVISNML  173
            GDG G+N+KVSAA++G  TC  D+   + VD S V NS   +N+CGI LVISNM+
Sbjct  120  GDGKGLNIKVSAAMEGPSTCEQDMADFK-VDPSAVKNSGDFQNICGIVLVISNMM  173


>SP:P83326 PMEI_ACTDE Pectinesterase inhibitor OS=Actinidia deliciosa OX=3627
GN=PMEI PE=1 SV=2
Length=185

 Score = 104 bits (260),  Expect = 2e-27
 Identities = 63/175 (36%), Positives = 98/175 (56%), Gaps = 7/175 (4%)
```

2. Prot-Scriber input

As input use sequence similarity search (Blast, Diamond, etc.) results

Swissprot and trEMBL

```
BLASTP 2.3.0+


Query= jg154936.t1

Length=353

>sp|Q9LFW1|RGP2_ARATH UDP-arabinopyranose mutase 2 OS=Arabidopsis thaliana OX=3702 GN=RGP2 PE=1 SV=1
Length=360

 Score = 676 bits (1745),  Expect = 1.50e-247
 Identities = 312/351 (88%), Positives = 334/351 (95%), Gaps = 3/351 (0%)

Query    3  SPTPLLKDELDIVIPTIRNLDFLEMWRPFFEPYHLIIVQDGDPAKTIKVPEGFDYELYNR 62
            +PTPLLKDELDIVIPTIRNLDFLEMWRPF +PYHLIIVQDGDP+K I VPEG+DYELYNR
Sbjct   13  NPTPLLKDELDIVIPTIRNLDFLEMWRPFLQPYHLIIVQDGDPSKKIHVPEGYDYELYNR 72

Query   63  NDINRILGRKANCISEKDSACRCFGYMVSKKKYIYTIDDDCFVANDRSGKQINALFQHIK 122
```

Local Sequence Alignment generated by Blast / Diamond

Lexical Analysis

Blast creates a special condition:

The more frequent a word the more informative it is.

- Exceptions are filtered out using regular expressions

lernwirkstatt-inklusion-oe.de

2. Algorithm sketch

1. Run sequence similarity searches against reference databases
2. Parse and filter using regular expression lists
3. Chop candidate descriptions into word-sets and score by
   a. Word-Frequency
4. Generate consensus descriptions as subsets of words
   a. Retain word order as in source candidate(s)
5. Assign highest scoring consensus description

2. Example query and its Blast results

Sprot

Probable mannitol dehydrogenase
8-hydroxygeraniol dehydrogenase
Geraniol dehydrogenase 1
Probable mannitol dehydrogenase 3
Cinnamyl alcohol dehydrogenase 8
Cinnamyl alcohol dehydrogenase 7
Mannitol dehydrogenase
Mannitol dehydrogenase
Probable mannitol dehydrogenase
Alcohol dehydrogenase 9
Probable cinnamyl alcohol dehydrogenase 9
Probable cinnamyl alcohol dehydrogenase 6
Probable mannitol dehydrogenase 1
Probable cinnamyl alcohol dehydrogenase 5
Probable cinnamyl alcohol dehydrogenase 3
Cinnamyl alcohol dehydrogenase 2
Cinnamyl alcohol dehydrogenase 3
Protein REDOX 1
Probable cinnamyl alcohol dehydrogenase 8B
Probable cinnamyl alcohol dehydrogenase 8C
Dehydroprecondylocarpine acetate synthase
Cinnamyl alcohol dehydrogenase 7
Probable cinnamyl alcohol dehydrogenase 9
Probable cinnamyl alcohol dehydrogenase 6

trEMBL

Uncharacterized protein
PKS_ER domain-containing protein
Uncharacterized protein
Putative mannitol dehydrogenase
uncharacterized protein LOC105165970
Putative alcohol dehydrogenase superfamily, zinc-type
Putative mannitol dehydrogenase
PKS_ER domain-containing protein
(thale cress) hypothetical protein
PKS_ER domain-containing protein
Alcohol dehydrogenase, C-terminal
PKS_ER domain-containing protein
probable mannitol dehydrogenase
PKS_ER domain-containing protein
Alcohol dehydrogenase
PKS_ER domain-containing protein
Alcohol dehydrogenase superfamily, zinc-type
Elicitor-activated gene 3-1
PKS_ER domain-containing protein
Cinnamyl-alcohol dehydrogenase family protein
PKS_ER domain-containing protein
Putative mannitol dehydrogenase
Cinnamyl-alcohol dehydrogenase
PKS_ER domain-containing protein

2. Filter out uninformative candidate descriptions

Using regular expressions, discard:

- Uncharacterized Protein
- Probable alcohol dehydrogenase
- Putative protein
- …

trEMBL

~~Uncharacterized protein~~
PKS_ER domain-containing protein
~~Uncharacterized protein~~
~~Putative mannitol dehydrogenase~~
~~uncharacterized protein LOC105165970~~
Putative alcohol dehydrogenase superfamily, zinc-type
~~Putative mannitol dehydrogenase~~
PKS_ER domain-containing protein
~~(thale cress) hypothetical protein~~
PKS_ER domain-containing protein
Alcohol dehydrogenase, C-terminal
PKS_ER domain-containing protein
~~probable mannitol dehydrogenase~~
PKS_ER domain-containing protein
Alcohol dehydrogenase
PKS_ER domain-containing protein
Alcohol dehydrogenase superfamily, zinc-type
Elicitor-activated gene 3-1
PKS_ER domain-containing protein
Cinnamyl-alcohol dehydrogenase family protein
PKS_ER domain-containing protein
~~Putative mannitol dehydrogenase~~
Cinnamyl-alcohol dehydrogenase
PKS_ER domain-containing protein

2. Phrases from candidate descriptions

```
[71] "c"
[72] "terminal"
[73] "alcohol c"
[74] "alcohol terminal"
[75] "dehydrogenase c"
[76] "dehydrogenase terminal"
[77] "c terminal"
[78] "alcohol dehydrogenase c"
[79] "alcohol dehydrogenase terminal"
[80] "alcohol c terminal"
[81] "dehydrogenase c terminal"
[82] "alcohol dehydrogenase c terminal"
```

Sample indices from reference description word-vector, maintaining order.

Source description

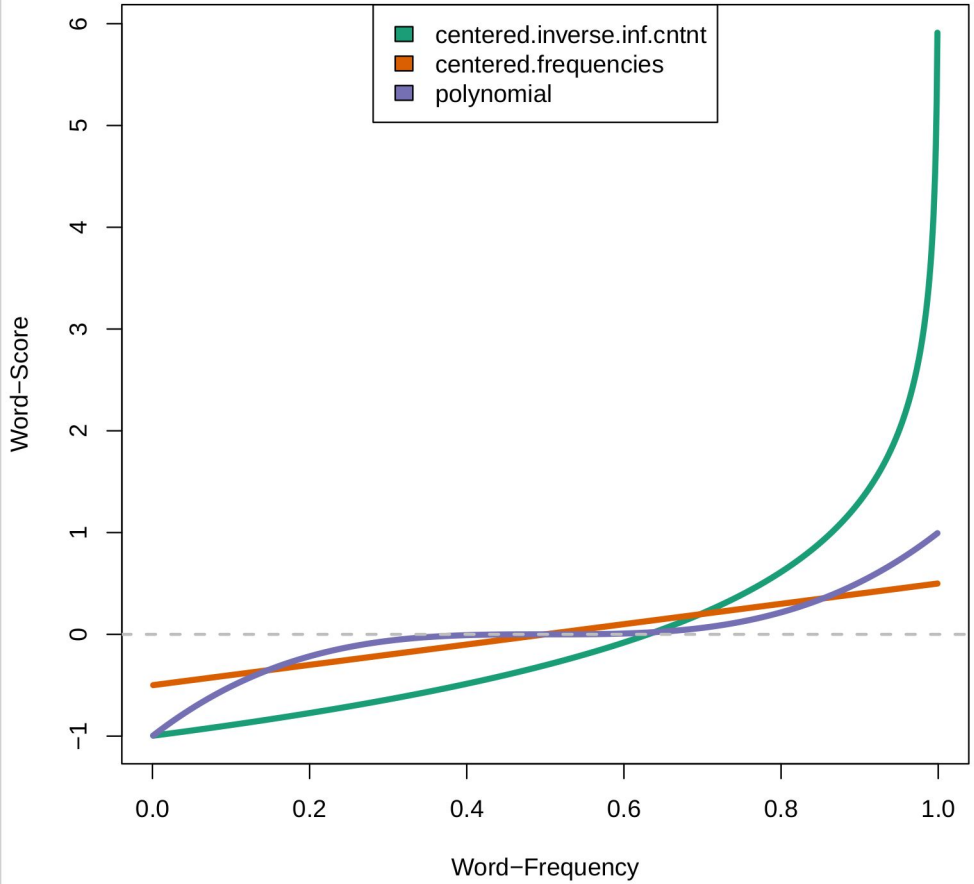
2. Phrases Scoring

General Approach:

- Filter out uninformative words
- Score each word depending on its frequency
- Penalize low frequency words -> Centering
- Apply function to phrase words (set of words) to compute phrase score.
  - Used here: `sum`

```
(?i)\bunknown\b
(?i)\bmember\b
(?i)\blike\b
(?i)\bassociated\b
(?i)\bcontaining\b
(?i)\bdomain\b
(?i)\bactivated\b
(?i)\bfamily\b
(?i)\bsubfamily\b
(?i)\binteracting\b
(?i)\bactivity\b
(?i)\bsimilar\b
(?i)\bproduct\b
(?i)\bexpressed\b
(?i)\bpredicted\b
(?i)\bputative\b
(?i)\bhypothetical\b
(?i)\buncharacterized\b
(?i)\bprobable\b
(?i)\bprotein\b
(?i)\bgene\b
(?i)\btair\b
(?i)\bfragment\b
(?i)\bhomolog\b
(?i)\bcontig\b
(?i)\brelated\b
(?i)\bremark\b
(?i)\b\w?orf(\w?|\d+)\b
(?i)\bof\b
(?i)\bor\b
(?i)\band\b
\b\d+\b
```

2. Word scoring functions in comparison



Word *w* (element of phrase *p*) scoring with functions of word frequency p_w:

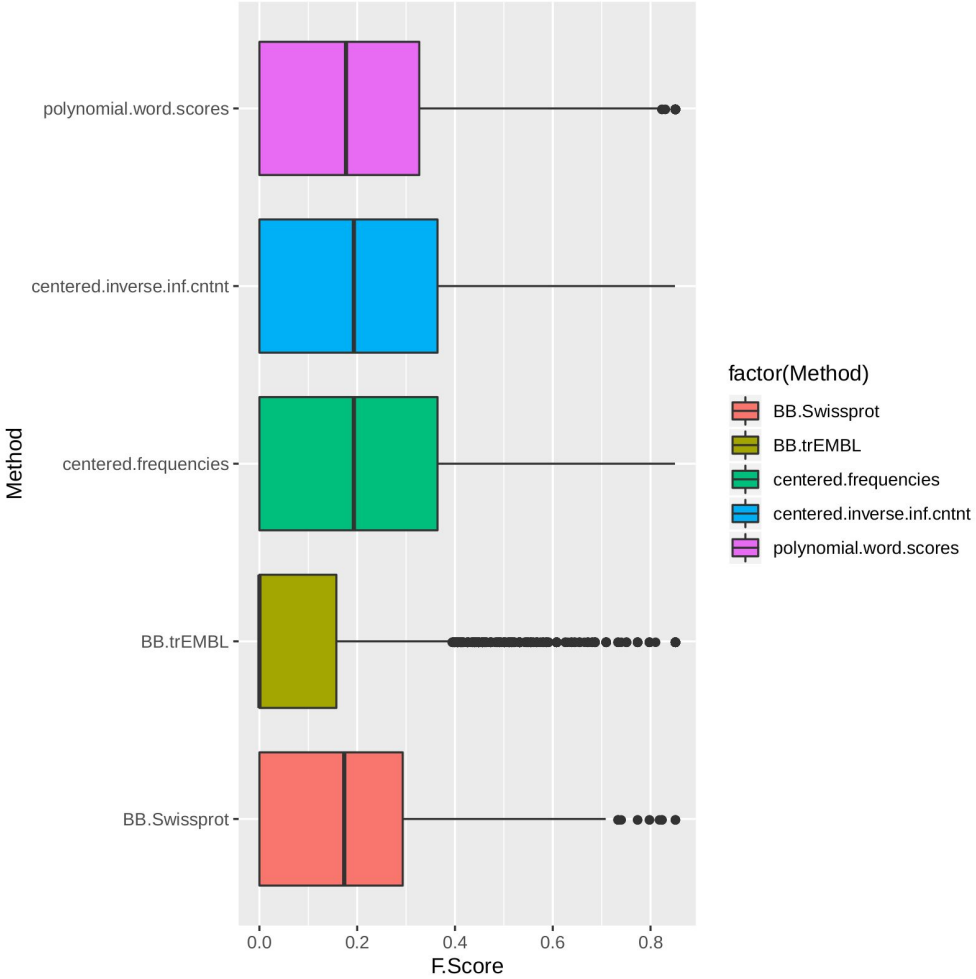
- $f(p_w) = log(1 - p_w) - mean(log(1 - p_{v_i \in p}))$
- $f(p_w) = p_w - \bar{p_w}$
- $f(p_w) = (2 * p_w - 1)^3$

2. Performance comparison: Best Blast (BB) vs Prot.Scriber

F-Score computed using words from reference annotations:

- Pfam-A (HMMER3)
- MapMan-Bins (Mercator)

Note that both Pfam-A and MapMan-Bin ontologies involve manual expert curation, and highly trustworthy annotation tools.



2. Prot.Scriber summary

- outperforms best Blast methods
- high throughput and fast protein function prediction method
- Rust implementation allows for high throughput
- algorithm works for gene family annotation: no other tool exists
- preparing publication: used in our last Nature paper on Fava Beans

README License

prot-scriber

Assigns short human readable descriptions (HRD) to query biological sequences using reference candidate descriptions. In this, `prot-scriber` consumes sequence similarity search (Blast or Diamond or similar) results in tabular format. A customized lexical analysis is carried out on the descriptions of these Blast Hits and a resulting HRD is assigned to the query sequences.

`prot-scriber` can also apply the same methodology to produce HRDs for sets of biological sequences, i.e. gene families.

Quick Start

(This section is for the lazy 😉 TL;DR)

`prot-scriber` can be used to generate human readable descriptions (HRDs) for *either* query biological *sequences* (proteins) or for *gene-families*. We will walk you through both use cases below with ready to use example input files.

Step 1 - Get `prot-scriber`

Depending on your operating system, download the ready to use executable from the table in section "Installation".

https://github.com/asishallab-group/prot-scriber

Thank you

h_da
hochschule
darmstadt